

MCP (Model Context Protocol)

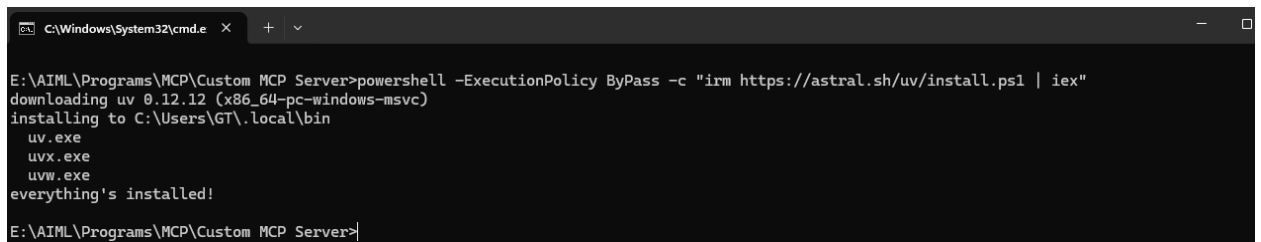
MCP Custom Tool & Claude Desktop Integration Guide

Step 1	Build a Custom Tool (MCP Server)
Step 2	Integrate with Claude Desktop (MCP Host)

Step 1: Build a Custom Tool (MCP Server)

1. Install uv and set up our Python project and environment (in Command Prompt).

```
powershell -ExecutionPolicy ByPass -c "irm https://astral.sh/uv/install.ps1 | iex"
```

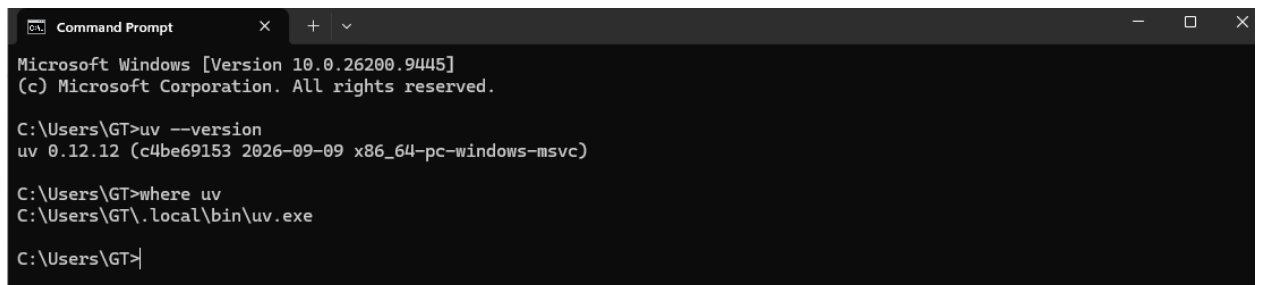


```
E:\AI\ML\Programs\MCP\Custom MCP Server>powershell -ExecutionPolicy ByPass -c "irm https://astral.sh/uv/install.ps1 | iex"
downloading uv 0.12.12 (x86_64-pc-windows-msvc)
installing to C:\Users\GT\local\bin
uv.exe
uvx.exe
uvw.exe
everything's installed!
E:\AI\ML\Programs\MCP\Custom MCP Server>
```

2. Close Command Prompt, then reopen Command Prompt.
3. Check the uv installation:

```
uv --version
```

```
where uv
```



```
Microsoft Windows [Version 10.0.26200.9445]
(c) Microsoft Corporation. All rights reserved.

C:\Users\GT>uv --version
uv 0.12.12 (c4be69153 2026-09-09 x86_64-pc-windows-msvc)

C:\Users\GT>where uv
C:\Users\GT\local\bin\uv.exe

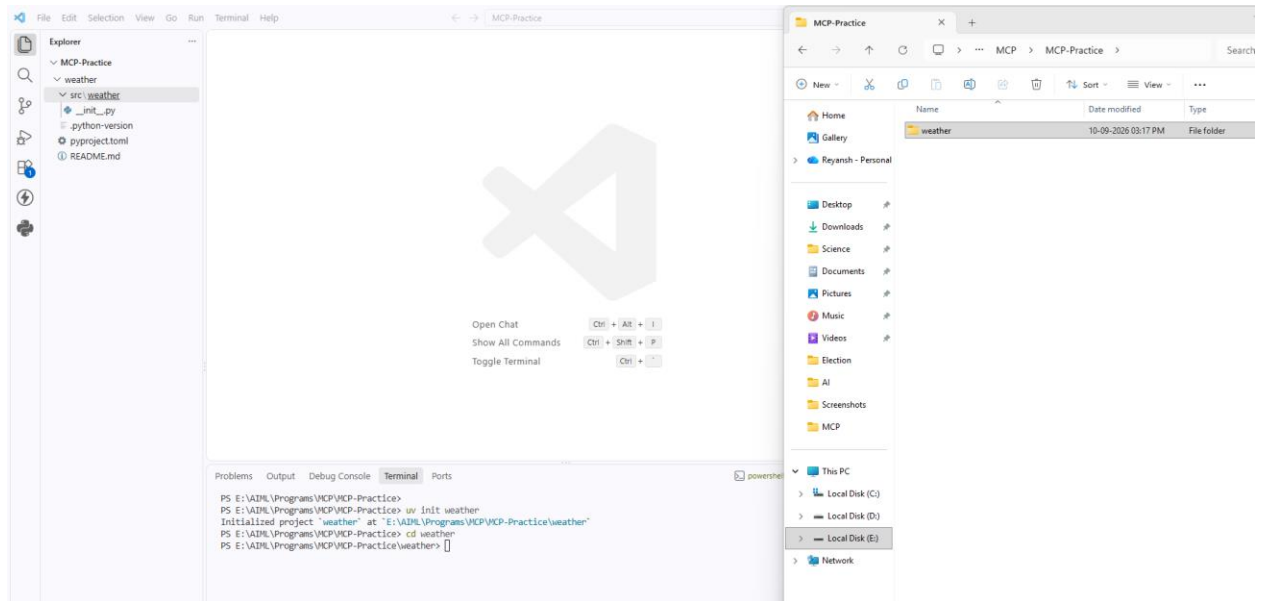
C:\Users\GT>
```

uv.exe location = "C:\Users\GT\local\bin\uv.exe"

4. Open VS Code. In terminal, create a new project "weather"

```
uv init weather
```

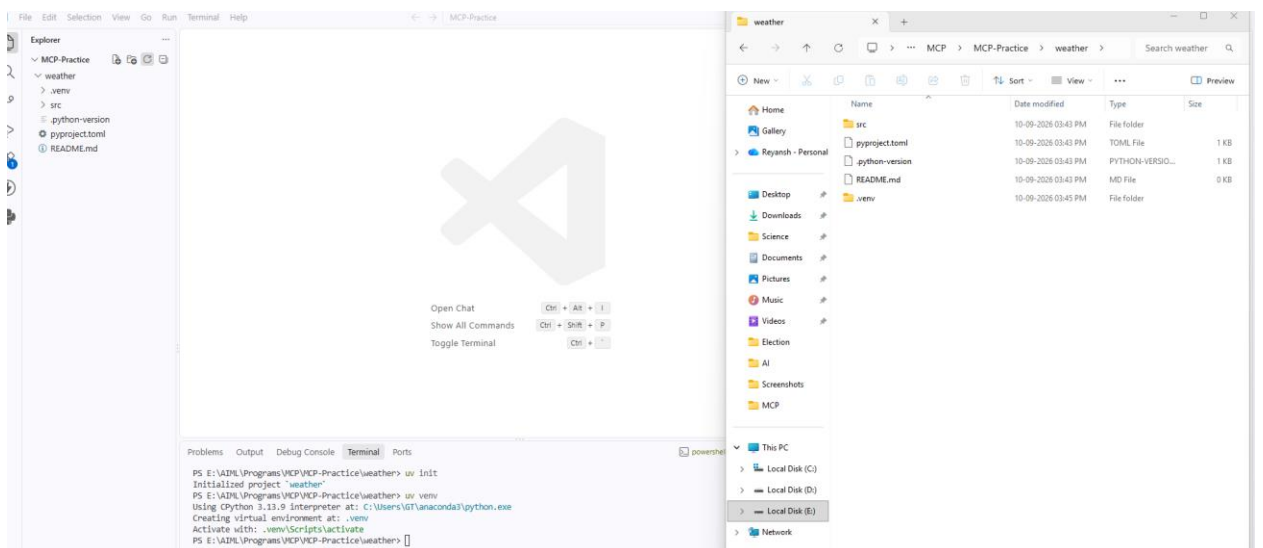
```
cd weather
```

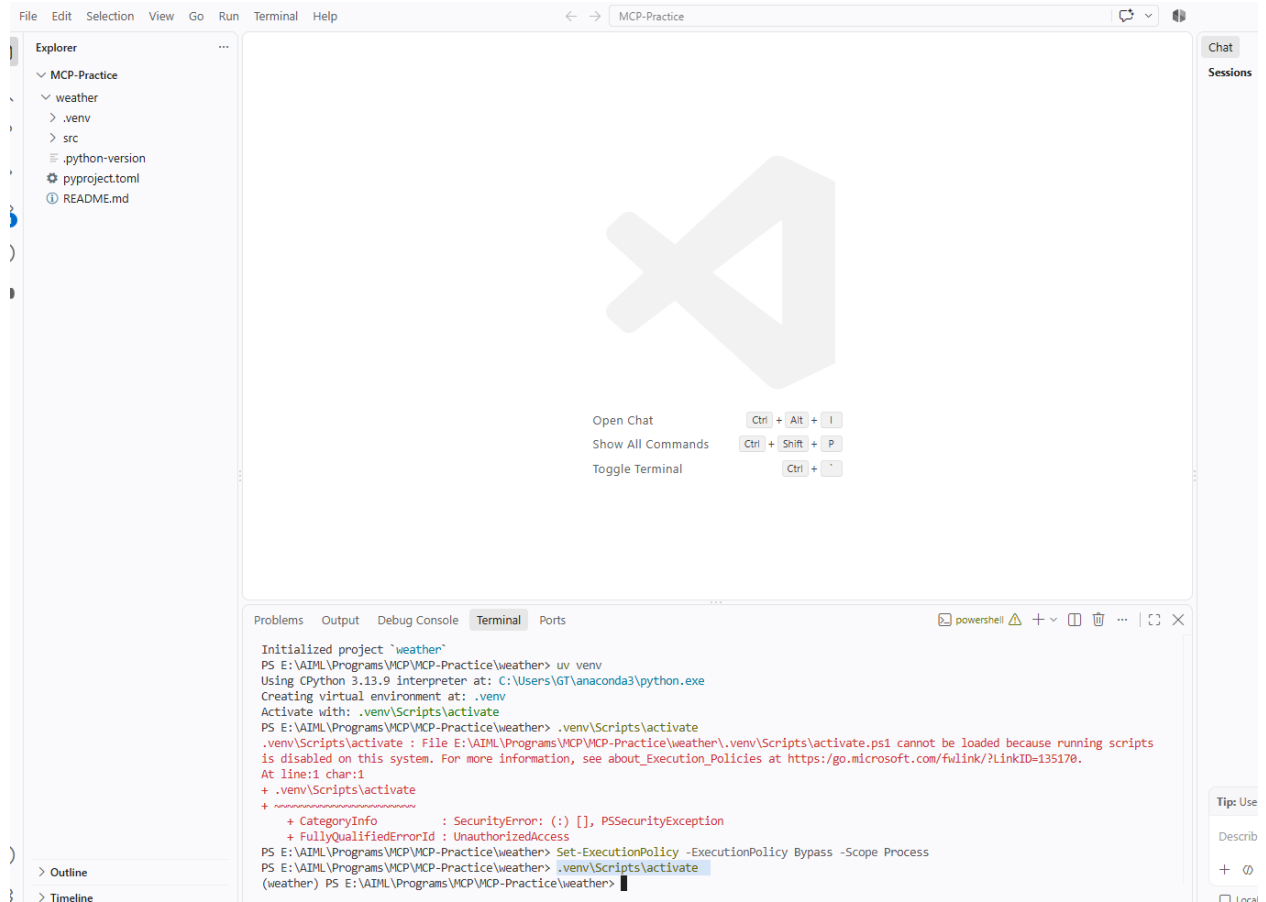


5. Create a virtual environment and activate it

`uv venv`

`.venv\Scripts\activate`





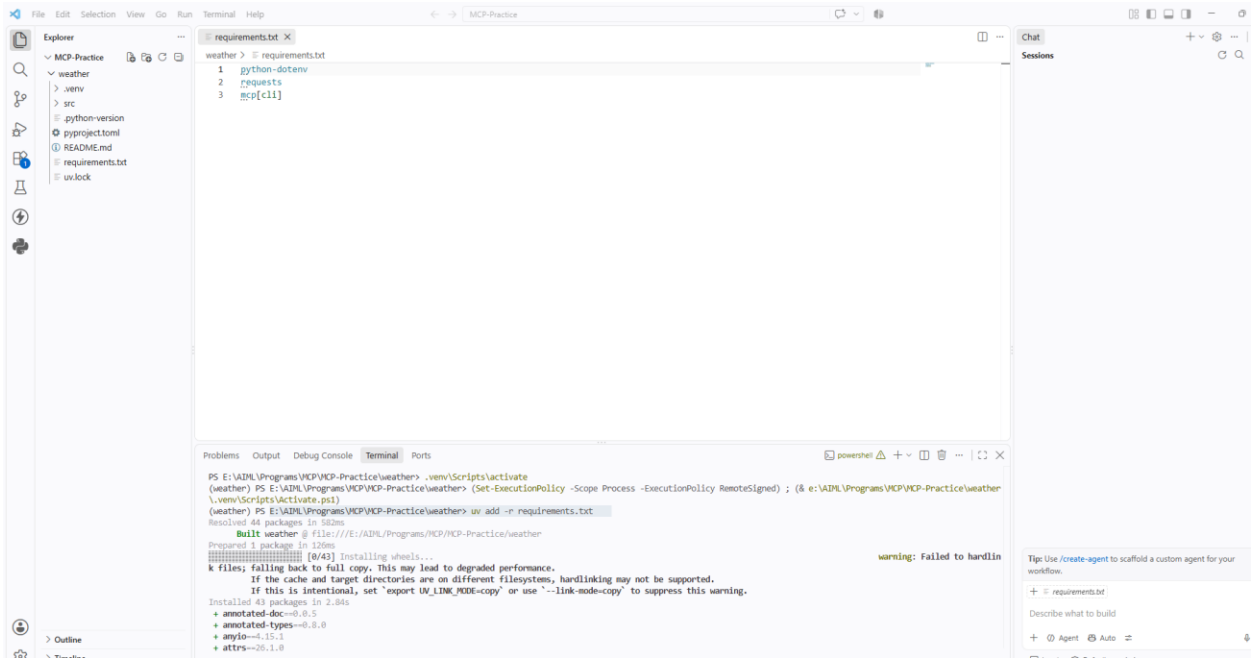
6. Install dependencies

Create Requirements.txt

mcp[cli]

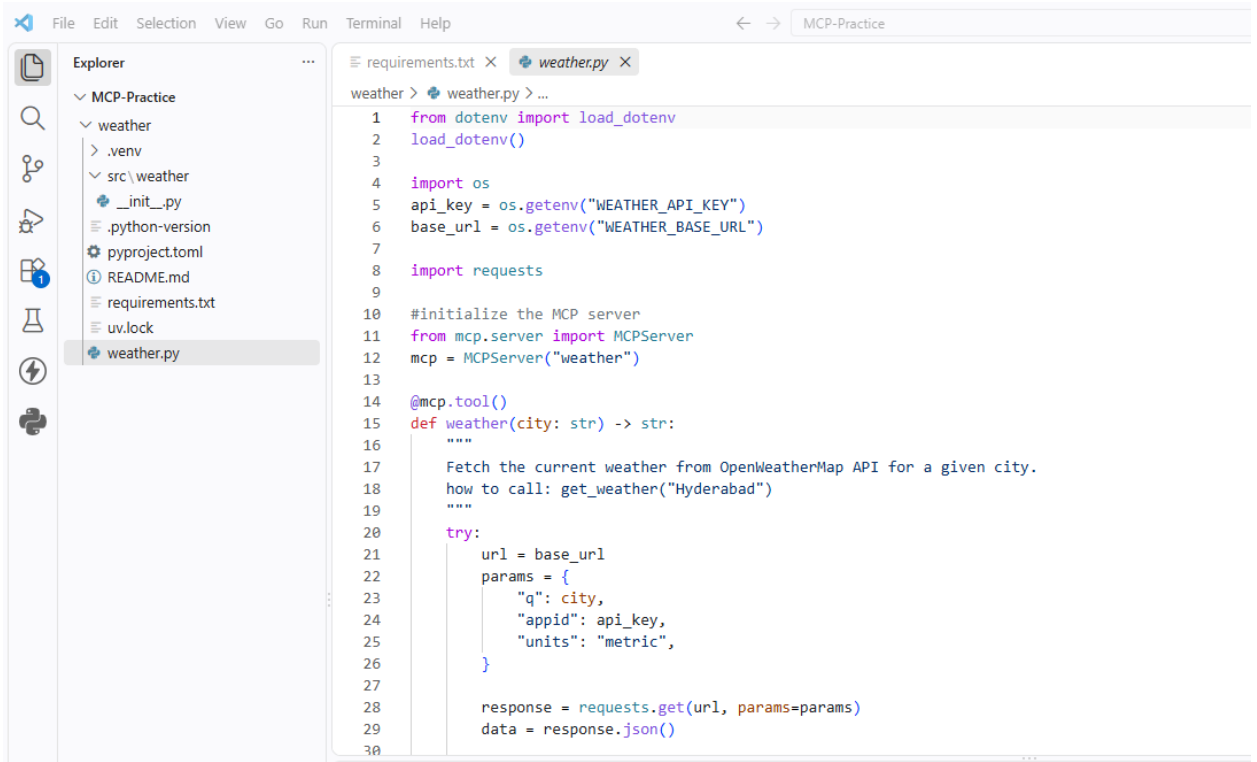
requests

uv add -r requirements.txt



7. Create Weather MCP tool – weather.py

<https://github.com/rajatray2106/AI ML/blob/main/Programs/MCP/Custom%20MCP%20Server/weather/weather.py>



8. Run the Python file to check for errors

uv run weather.py

File Edit Selection View Go Run Terminal Help MCP-Practice

Explorer

- MCP-Practice
 - weather
 - .venv
 - src\weather
 - __init__.py**
 - .python-version
 - pyproject.toml
 - README.md
 - requirements.txt
 - uv.lock
 - weather.py**

requirements.txt X weather.py X

weather > weather.py > ...

```
1 from dotenv import load_dotenv
2 load_dotenv()
3
4 import os
5 api_key = os.getenv("WEATHER_API_KEY")
6 base_url = os.getenv("WEATHER_BASE_URL")
7
8 import requests
9
10 #initialize the MCP server
11 from mcp.server import MCPServer
12 mcp = MCPServer("weather")
13
14 @mcp.tool()
15 def weather(city: str) -> str:
16     """
17     Fetch the current weather from OpenWeatherMap API for a given city.
18     how to call: get_weather("Hyderabad")
19     """
20     try:
21         url = base_url
22         params = {
23             "q": city,
24             "appid": api_key,
25             "units": "metric",
26         }
27
28         response = requests.get(url, params=params)
29         data = response.json()
30
```

Problems Output Debug Console Terminal Ports

+ requests==2.34.2
+ rich==15.0.0
+ rpsd-py==2026.6.3
+ shellingham==1.5.4
+ sse-starlette==3.4.11
+ starlette==1.6.0
+ truststore==0.10.4
+ typer==0.27.2
+ typing-extensions==4.16.0
+ typing-inspection==0.4.4
+ urllib3==2.7.0
+ uvicorn==0.52.4
+ weather==0.1.0 (from file:///E:/AI/ML/Programs/MCP/MCP-Practice/weather)

(weather) PS E:\AI\ML\Programs\MCP\MCP-Practice\weather>
(weather) PS E:\AI\ML\Programs\MCP\MCP-Practice\weather> uv run weather.py

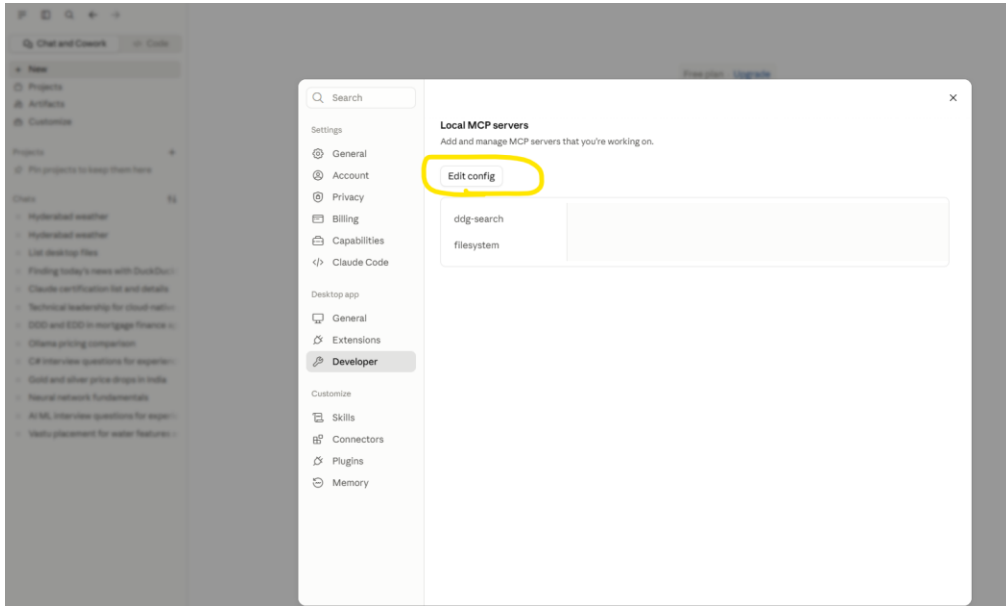
> Outline
> Timeline

Step 2: Integrate with Claude Desktop (MCP Host)

1. Install Claude Desktop

2. Open the Claude Configuration File

Open your Claude Desktop application → File → Settings → Developer Options → click Edit Config. This opens your `claude_desktop_config.json` file in your default text editor.



3. Copy & Paste the MCP server configuration

```
{
  "mcpServers": {
    "weather": {
      "command": "C:\\Users\\GT\\.local\\bin\\uv.exe",
      "args": [
        "run",
        "--directory",
        "E:\\AIML\\Programs\\MCP\\Custom MCP Server\\weather",
        "weather.py"
      ]
    }
  }
}
```

```

1 {
2   "mcpServers": {
3     "weather": {
4       "command": "C:\\Users\\GT\\.local\\bin\\uv.exe",
5       "args": [
6         "run",
7         "--directory",
8         "E:\\AIML\\Programs\\MCP\\Custom MCP Server\\weather",
9         "weather.py"
10      ]
11    },
12    "ddg-search": {
13      "command": "uvx",
14      "args": [
15        "duckduckgo-mcp-server"
16      ]
17    },
18    "filesystem": {
19      "command": "npx.cmd",
20      "args": [
21        "-y",
22        "@modelcontextprotocol/server-filesystem",
23        "C:\\Users\\GT\\Desktop",
24        "C:\\Users\\GT\\Downloads"
25      ]
26    }
27  },
28  "coworkUserFilesPath": "C:\\Users\\GT\\Claude",
29  "preferences": {
30    "coworkBrowserToolsEnabled": false,
31    "coworkPreferredBrowser": "built_in",
32    "launchPreviewPersistedWorkspaces": [],
33    "launchPreviewSessionScopedSessions": [],
34    "coworkLegacyRootGrantsPruned": true,
35    "coworkScheduledTasksEnabled": false,
36    "coworkHippaRestricted": false,
37    "orgWorkAcrossAppsDisabled": true,
38    "ddg-search-max-results": 10
39  }
40 }

```

4. Close Claude desktop instance. Make sure to close its all instances from Taks Manager.

Task Manager

Search: Type a name, publisher, or PID...

Processes

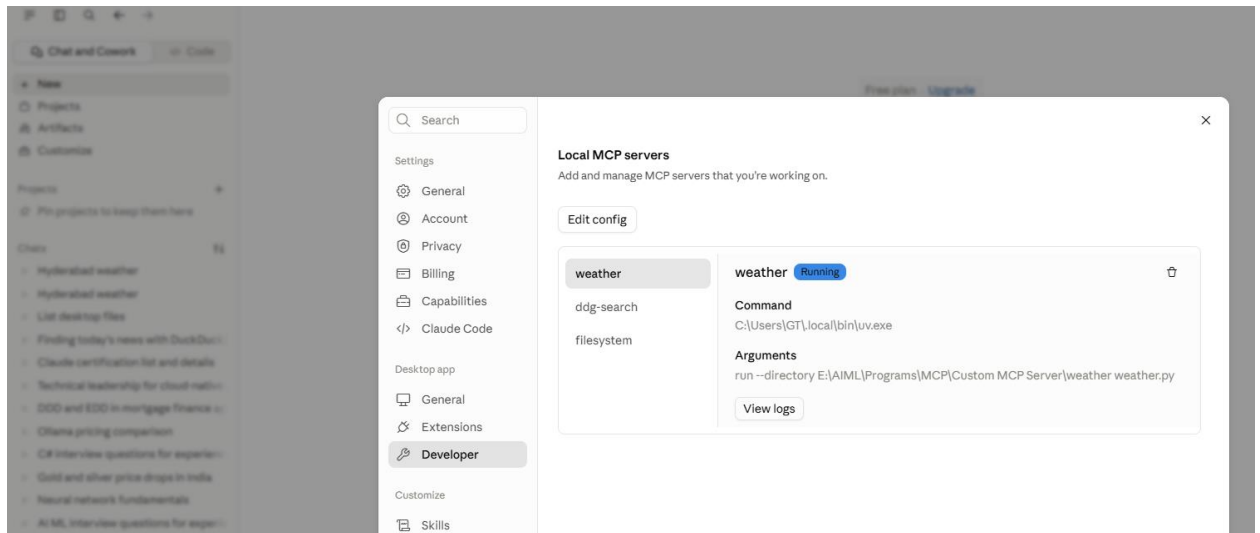
Run new task End task Efficiency mode

Name	Status	CPU	Memory	Disk	Network
Google Chrome (18)		4.5%	943.2 MB	0.1 MB/s	0 Mbps
Claude (9)		0.4%	321.9 MB	0 MB/s	0.1 Mbps
Visual Studio Code (13)		0%	289.6 MB	0.1 MB/s	0 Mbps
Snipping Tool		0%	132.2 MB	0 MB/s	0 Mbps
Search (7)		0%	118.4 MB	0 MB/s	0 Mbps
Windows Explorer		5.3%	86.7 MB	0 MB/s	0 Mbps
Start		0%	76.6 MB	0 MB/s	0 Mbps
McAfee Framework Host Servi...		0%	73.6 MB	0.1 MB/s	0 Mbps

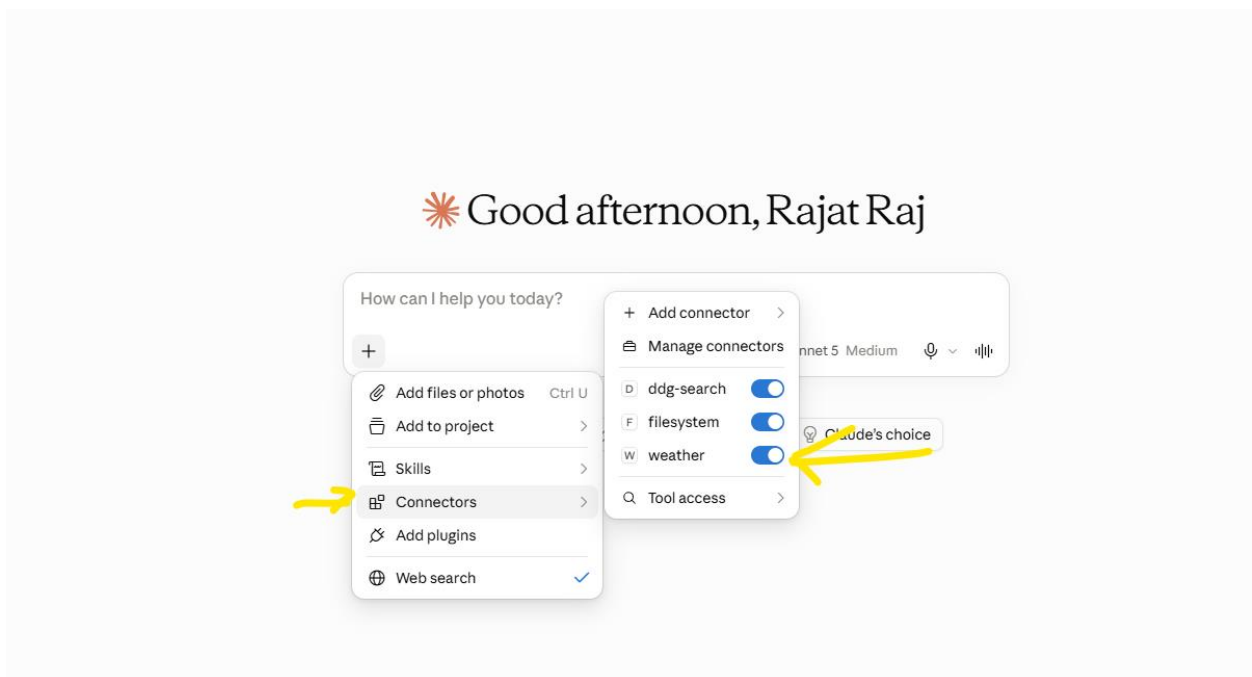
5. Open a new instance of Claude desktop.

6. Go to

File -> Settings -> Developer -> A new entry for weather MCP server will appear here.



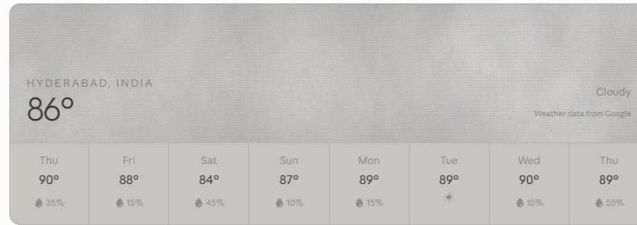
7. Validate connector



7. Search for weather in Claude desktop

Hyderabad weather ▾

weather("Hyderabad")



Hyderabad is currently cloudy at 86°F. Today's high is 89.5°F with a 35% chance of rain. Rain chances stay elevated through Saturday (45%) before clearing up mid-week, then pick back up to 55% next Thursday.

just now

use weather mcp, how is weather in Mumbai

Loaded tools, used weather integration >

Mumbai is currently seeing broken clouds at around 31°C (88°F).

just now



+ Write a message...



Please do not make mistakes. Please do not make mistakes.

Connet 5. Medium

Hyderabad weather ▾

use weather mcp, how is weather in Mumbai

Loaded tools, used weather integration >

Mumbai is currently seeing broken clouds at around 31°C (88°F).

is it coming from weather mcp?

Yes — that Mumbai reading came from the `weather:weather` MCP tool (OpenWeatherMap under the hood), not the same `weather_fetch` tool I used for Hyderabad. It only returns current conditions (no forecast), which is why that response was shorter.

just now



How to deploy our MCP server to marketplease?

- Use Smithery.ai
- Smithery.ai is a marketplace and infrastructure platform for MCP servers.